

TL Corp CRM Portal v12.0

FULL Functional + Technical Specification (SRD/PRD Hybrid)

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Purpose: Provide a no-ambiguity, build-ready specification for the TL Corp Digital Operating System. This document defines required behavior down to screen, button, validation, event, and audit level, and includes a recommended technical architecture (data model + API expectations) suitable for replacing paper workflows, supporting dispute defense, and scaling into commercial/enterprise use.

Prototype reference: crm-tl.vercel.app

0. Non■Negotiables (Definition of Success)

- Post-login behaves like an app (SPA): instant transitions, persistent navigation, no full page reloads.
- RBAC enforced server-side: clients see only their jobs; employees see only assigned jobs; admins control all.
- 3■Tap Rule for field actions from Employee Today: Navigate, Open Job, Clock In/Out, Add Photo, Add Task.
- Auto-save drafts everywhere + Resume Draft.
- Audit trail is immutable for critical events (schedule/scope/approvals/publishing/deletes).
- Electronic signatures with version lock + signed PDF snapshot.
- Document tracking events: Sent/View/Sign with notifications; events tied to document version.
- Replace paper-based workflows completely (daily logs, approvals, closeout packets).
- Usability: no user should need to call to learn flows; in-app help + tooltips + microcopy required.

1. Product Scope & Portal Breakdown

One system, one database, three portal shells: Employee, Client, Admin. Each role gets a tailored UI/permissions view of the same underlying data.

- Employee Portal (mobile-first): Today/Week, Job Bible (Tasks/Photos/Docs/Daily Log/Time/Messages), Fast field capture, acknowledgments.
- Client Portal (calm + curated): Overview, Updates, Photos (published only), Documents, Change Orders, Billing/Payments, Messages.
- Admin Portal (desktop-first): Dashboard, Jobs, Schedule/Dispatch, Publish Queue, COs, Invoices, Users/Roles, Templates, Reports.

2. Branding + Design System (Tokens are mandatory)

Implement tokens once and consume them across all components. Never hardcode per-component colors/spacing.

- Logo: fixed top-left in app shell; never stretched; maintain aspect ratio; clearspace $\geq$ height of the 'T' in TL mark.
- Primary TL Blue: #0B2A4A | Accent Gold: #C7A252 | Success #16A34A | Warning #F59E0B | Danger #DC2626
- Typography: Inter preferred; fallback system-ui.
- Spacing: 8pt grid (4/8/12/16/24/32/48/64).
- Tap targets: 44×44px minimum; button height 44–48px; input height 44px.
- Component set: AppShell, TopBar, Card, Button variants, Badge, Modal, Drawer, Toast, Skeleton, EmptyState, Stepper/Wizard, FileUpload, Table.

2.1 UI State Matrix (every component)

- Default, Hover (desktop), Pressed, Focus ring (2px #60A5FA), Disabled, Loading, Error, Success (where applicable).
- Every primary action must produce: visible state change + success toast + (if relevant) audit event.

3. Recommended Technical Architecture

This defines a reference architecture your dev can implement with Next.js/React + API routes/server + Postgres + object storage. The key is: server-side RBAC, immutable audit logging, and reliable event tracking.

- Frontend: SPA shell (Next.js app router or similar), role-based route guards; responsive layout.
- Backend: REST or GraphQL API; server-side authorization; background jobs for notifications (queue).
- Database: PostgreSQL (preferred) with UUID primary keys; row-level access enforced in service layer.

- Files: object storage (S3-compatible) for photos/docs; generate thumbnails; store metadata in DB.
- Search: simple DB search initially; optional full-text later.
- Observability: request logging + error tracking; audit export endpoints.
- Security: HTTPS only; secure sessions; optional 2FA for admin.

4. Data Model (Core Tables / Objects)

Minimum viable data model that supports all workflows. Field names below are canonical and should remain stable.

- User, Role, Permission, UserProjectAccess
- Project (Job), ProjectClientLink, Assignment (ScheduleItem), Milestone
- Task, TaskComment, TaskAttachment
- Photo, PhotoComment, PhotoPublishRequest
- Document, DocumentVersion, DocumentRecipient, DocumentEvent
- DailyLog, DailyLogSection, DailyLogAttachment
- TimeEntry, TimeEntryEditRequest
- ChangeOrder, ChangeOrderVersion, ChangeOrderSignature
- Invoice, Payment, Receipt
- MessageThread, Message, MessageAttachment
- AuditLog (immutable)

4.1 Project (Job) schema (required fields)

Field	Requirement
id	UUID
job_number	String; unique; format configurable (e.g., 26-014)
name	String; required
type	Enum: Residential / Commercial / Maintenance / Service / Warranty
status	Enum lifecycle: Draft→Scheduled→Assigned→Acknowledged→In Progress→Pending Review
address	Structured: street/city/state/zip + geo lat/long
access_notes	Text; optional
work_hours_allowed	Text/structured; optional
scope_summary	Text; required (admin)
today_target	Text; optional (employee-facing)

safety_notes	Text; optional
lead_user_id	FK User; required once Scheduled
created_by	FK User; required
created_at/updated_at	Timestamps

4.2 Assignment (ScheduleItem) schema

Field	Requirement
id	UUID
project_id	FK Project; required
start_at	Datetime; required
duration_minutes	Int; required
assigned_user_ids	Array FK Users; required
status	Enum: Draft / Assigned / Acknowledged / Completed
priority	Enum: Normal / Urgent / Emergency
critical_note	Short text shown on Today card; optional
changed_since_last_login_flag	Computed per user; drives banner
created_by/updated_by	FK User
created_at/updated_at	Timestamps

5. Roles & Permissions (RBAC)

RBAC must be enforced server-side. This table defines minimum permission expectations. 'R' = read, 'C' = create, 'U' = update, 'D' = delete, 'P' = publish.

Module	Client	Employee	Lead/Foreman	Admin
Projects/Jobs	R (own)	R (assigned)	R (assigned)	CRUD
Assignments/Schedule	R (own)	R (assigned)	R (assigned)	CRUD
Tasks	R (published opt.)	CRU (assigned)	CRU (assigned)+Assign	CRUD+Verify
Photos	R (published)	CR (internal)	CR (internal)	CRUD+Publish
Docs	R (published)	R (assigned)	R (assigned)	CRUD+Publish
Daily Logs	R (published summary opt.)	CRU (own draft)	CRU+Submit	CRUD+Export
Time	N/A	CRU (own)	CRU (crew view opt.)	CRUD+Approve
Change Orders	CRU (approve/decline)	R	R	CRUD+Send
Invoices/Payments	R+Pay	N/A	N/A	CRUD
Messages	R/W (admin thread)	R/W (job thread)	R/W	CRUD+Moderate
Audit/Reports	R (own exports opt.)	N/A	N/A	R+Export

6. Notification Engine (Required)

All important events generate in-app notifications. Email/SMS are optional channels. Notifications must deep-link to the exact object.

- Channels: In-app bell (required), Email (configurable), SMS (phase 2).
- Anti-spam: view events notify once per recipient (first view); subsequent views roll into activity feed or batched summary.
- Types: Assignment created/changed, Task assigned/blocked, Publish request, Doc viewed, Doc signed, CO approved/declined, Invoice paid, Daily log submitted, Safety acknowledgment.

6.1 Document tracking events (canonical)

- Created, Sent, Delivered(optional), Opened(optional email), Viewed(viewer), Downloaded(optional), Signed/Approved, Declined, Expired, Revoked.

6.2 AuditLog events (immutable)

- Schedule changes (old/new), job status changes, task status changes, publish actions, signature events, deletions/archives, reopen log with reason.

7. Employee Portal — Screen Specifications

All employee screens must be optimized for phones/iPads, low typing, big actions. Every screen includes a one-line 'What to do here' header and Help icon.

7.1 E1 Today (Default)

- Loads assignments for Today, sorted by start time.
- Job card fields: start time, job name/job#, address (truncate), crew, priority tags, critical note (1 line).
- Primary buttons per card: Navigate, Open Job, Clock In.
- Banner if updates since last login; tap opens Change Summary drawer listing changed fields.
- Long-press card opens quick actions: Add Photo, Add Task, Message Admin, View Docs.
- Empty state: 'No assigned work today' + buttons View My Week + Message Admin.

7.2 E2 My Week

- 7-day list, Today expanded by default; expand/collapse per day.
- Swipe assignment: Navigate, Message Admin.
- Filter toggle: All / Urgent / Emergency.

7.3 E3 Job Bible (Shell)

- Sticky Critical Info panel: Address (Navigate), Access notes, Scope summary, Today target, Safety/PPE, Contacts (tap-to-call).
- Tabs max 6: Tasks, Photos, Docs, Daily Log, Time, Messages.
- Back preserves prior scroll position.
- More menu: Request Materials, Report Issue, View Change Summary.

7.4 E4 Tasks (Tab)

- Sections: Open, In Progress, Blocked, Completed (collapsed).
- CTA: + New Task.
- Swipe left = Complete (confirm), swipe right = Block (confirm).
- Block requires blocked_reason dropdown + optional note/photo.
- Complete requires photo for categories QA/Safety/Punch (configurable rule).

7.5 E5 Task Detail

- Editable fields (if permitted): status, assignee, due date, notes, attachments.
- Comment thread with @mentions.
- Mini audit display: created by/at, last updated by/at.
- Actions: Duplicate (lead/admin), Cancel (lead/admin), Delete (time-window rule).

7.6 E6 Photos (Tab)

- + Photo opens camera; post-capture requires Tag (Before/During/After).
- Visibility defaults Internal; employee can Request Publish (creates admin queue item).
- Viewer supports comment; delete rules: uploader within 10 min or admin; Undo 10 min.

7.7 E7 Docs (Tab)

- Folders: Scope/Contract, Plans, Specs, Permits, Safety, Other.
- Latest versions default; toggle All versions read-only.
- PDF viewer: pinch zoom; optional search (phase 2).

7.8 E8 Daily Log (Tab)

- Status: Not started / Draft / Submitted.
- Sections: Work performed (template checkboxes + notes), Deliveries, Issues/Blocks, Safety, Client interaction.
- Unlimited photo attachments, tagged per section.
- Auto-save every 10 seconds; Submit locks; admin reopen requires reason (audit).
- If Issues entered → prompt: 'Create task from this?'

7.9 E9 Time

- Clock in/out by job; show running timer; Switch Job stops current and starts new (logs both).
- Break prompt at clock-out if missing.
- Edit requests allowed; admin approves.

7.10 E10 Messages

- Job thread + task comment threads; attachments internal unless published.

8. Client Portal — Screen Specifications

Client portal must be calm, curated, and non-technical. Clients see only published items.

8.1 C1 Overview

- Project status, milestone timeline, next steps.
- Action Items: Approve CO, Pay Invoice, Message Admin (conditional).
- Latest update preview.

8.2 C2 Updates

- Weekly/date cards; update detail includes summary + published photos + linked docs.

8.3 C3 Photos

- Published photos only; albums Before/During/After; viewer with optional download.

8.4 C4 Documents

- Contracts/estimates/CO PDFs; view-only by default; download configurable.

8.5 C5 Change Orders

- Pending first; CO detail shows scope/cost/time + disclaimer.
- Approve: e-sign typed+drawn; Decline requires reason; audit logged.

8.6 C6 Billing

- Invoice list + detail; Pay; receipt download; payment event logged.

8.7 C7 Messages

- Client↔Admin only; topic selector for new messages.

9. Admin Portal — Screen Specifications

Admin portal is the control center: job creation, scheduling, approvals, publishing, reporting.

9.1 A1 Dashboard

- Widgets: Today jobs, blocked tasks, publish requests, pending CO approvals, unpaid invoices, crew utilization.
- Widgets deep-link to filtered views.

9.2 A2 Jobs List

- Filters: status/type/lead/date range; row actions: Open, Schedule, Publish, Archive.
- Never hard-delete projects; Archive only.

9.3 A3 Create Project Wizard

- Step 1 Client+Address+Type; Step 2 Scope+Access+Hours+Safety; Step 3 Template; Step 4 Schedule+Crew; Step 5 Client visibility; Finish.
- Validation prevents missing required fields; finish triggers notifications.

9.4 A4 Schedule/Dispatch

- Create assignment; change assignment triggers employee 'Updated since last login' banner; optional acknowledgment required.

9.5 A5 Publish Queue

- Queues: photo publish requests, doc publish requests, update drafts; publish logs publisher+timestamp.

9.6 A6 Change Orders

- Create CO with scope/cost/time/disclaimer; send; track view/sign; approved version locked; optional invoice generation.

9.7 A7 Invoices & Payments

- Create/send invoices; track view/pay; receipt stored; audit logged.

9.8 A8 Users/Roles

- Invite users; assign roles; project access; disable retains audit history.

9.9 A9 Reports/Exports

- Dispute Defense Packet export (contract/COs/logs/photos/audit).
- Audit log export per project.
- Daily log export with embedded photos.

10. Legal Grade Workflows (Paper Replacement)

These workflows are required to make the portal legally defensible and commercial-ready.

10.1 Signature capture requirements

- Signature types: typed name + drawn signature; optional signature image upload.
- Metadata: signer name/email/role, timestamp, IP (optional), device info/user-agent (optional).
- After signature: generate locked PDF snapshot; store in immutable archive folder; version lock.

10.2 Daily log legal standard

- Unlimited photos; each stamped with user+timestamp; logs locked on submit; reopen requires reason + audit.
- Export PDF includes log text + embedded photos + timestamps.

10.3 Closeout & signoff

- Completion gate: required tasks/photos/checklists must be satisfied before marking Complete.
- Optional client on-site completion signature; warranty acknowledgment signature; punch signoff.

11. API Expectations (Reference Contract for Dev)

Developer may choose REST or GraphQL. Below is a recommended REST shape to eliminate ambiguity.

- **POST /auth/login | POST /auth/logout | POST /auth/register**
- **GET /me (current user + role + permissions)**
- **GET /projects?filters=... | POST /projects | GET /projects/:id | PATCH /projects/:id | POST /projects/:id/archive**
- **POST /assignments | GET /assignments/today | GET /assignments/week | PATCH /assignments/:id**
- **POST /tasks | GET /projects/:id/tasks | PATCH /tasks/:id | POST /tasks/:id/block | POST /tasks/:id/complete**
- **POST /photos (upload) | GET /projects/:id/photos | POST /photos/:id/publish-request | POST /photos/:id/publish**
- **POST /daily-logs/start | PATCH /daily-logs/:id | POST /daily-logs/:id/submit | POST /daily-logs/:id/reopen**
- **POST /time-entries/clock-in | POST /time-entries/clock-out | POST /time-entries/switch-job**

- **POST /change-orders | POST /change-orders/:id/send | POST /change-orders/:id/approve | POST /change-orders/:id/decline**
- **POST /invoices | POST /invoices/:id/send | POST /payments (webhook)**
- **GET /documents/:id/viewer (tokenized) | GET /documents/:id/events**
- **GET /audit?project_id=... | GET /exports/dispute-packet?project_id=...**

12. Acceptance Test Scenarios (QA Checklist)

These are executable test cases. Build is not complete until all pass on iPhone, iPad, and desktop.

- EMP-01: Employee opens app → Today loads within 2s → sees correct assignments.
- EMP-02: Employee clocks in from Today card → timer starts → toast appears → time entry created.
- EMP-03: Employee adds task → blocks it with reason → admin sees blocked task → audit entry exists.
- EMP-04: Employee submits daily log with photos → log locks → export PDF contains embedded photos.
- CLI-01: Client receives contract link → opens viewer → admin receives 'Viewed' alert.
- CLI-02: Client signs contract/CO → signed PDF snapshot stored → admin receives 'Signed' alert.
- ADM-01: Admin creates project via wizard → assigns schedule → employees notified → 'Updated since last login' banner works.
- ADM-02: Admin publishes photos → client sees them immediately, internal photos remain hidden.
- ADM-03: Dispute packet export generates bundle with contract, COs, logs, photos, audit timeline.
- SEC-01: Client cannot access another client's project via URL manipulation (RBAC enforced).

13. Delete/Archive Rules (Controlled Destruction)

To preserve legal defensibility, most objects are archived/cancelled rather than deleted.

- Projects: never hard-delete; Archive only.
- Tasks: creator may delete within 10 minutes; after that admin only; otherwise mark Cancelled (kept in audit).
- Photos: uploader may delete within 10 minutes; otherwise admin only; Undo 10 minutes.
- Docs: never delete versions; new upload creates new version; old versions remain read-only.
- Messages: never delete; allow admin redaction only with audit note (rare).